

## Lab 3: Component Modelling & Architectural Pattern Selection

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| <b>Section</b>       | H                                                                                                                                                             |
| <b>Lab</b>           | Lab 3 – Component Modelling & Architectural Pattern Selection                                                                                                 |
| <b>Scenario used</b> | Problem Statement #18: Clinical Trial Subject Recruitment Engine (as per Lab 1 problem-statement PDF, used here in place of the sample Coffee Kiosk scenario) |

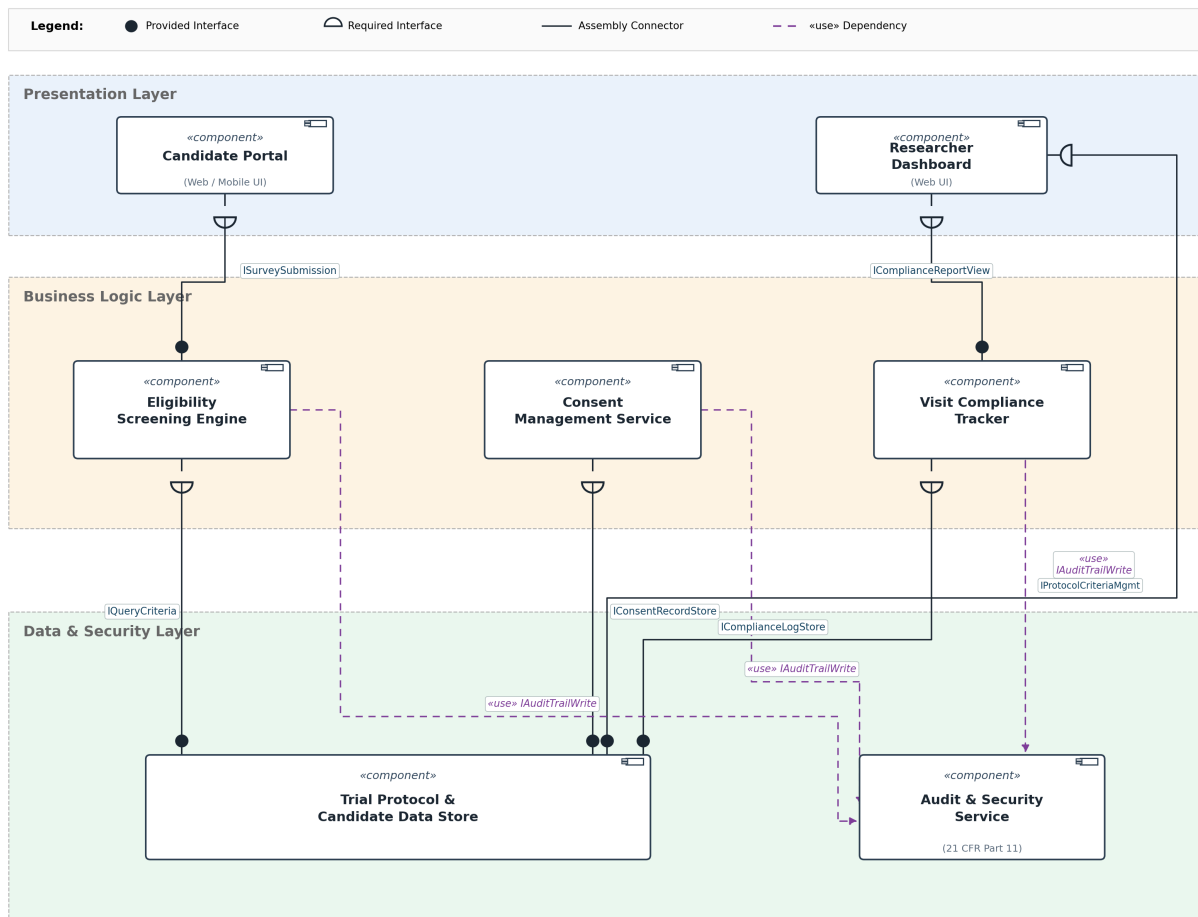
### Scenario Note

This submission applies the Lab 3 component-modelling and architecture-selection workflow to the system described in Problem Statement #18 (Lab 1): a Clinical Trial Subject Recruitment Engine for medical research institutions. The platform performs eligibility pre-screening of trial candidates against protocol inclusion/exclusion criteria, tracks informed-consent milestones, and logs visit compliance, while meeting 21 CFR Part 11 electronic-record and audit-trail requirements. The two stakeholders are the Trial Candidate and the Clinical Researcher.

### 1. Component Diagram

Architectural style: Layered Architecture (Presentation → Business Logic → Data & Security).

## UML Component Diagram — Clinical Trial Subject Recruitment Engine (Layered Architecture)



### Components (7)

- Candidate Portal – web/mobile UI where candidates submit screening survey responses (Presentation Layer).
- Researcher Dashboard – web UI for researchers to manage protocol criteria and review compliance (Presentation Layer).
- Eligibility Screening Engine – evaluates candidate responses against inclusion/exclusion criteria (Business Layer).
- Consent Management Service – tracks informed-consent milestones (Business Layer).
- Visit Compliance Tracker – logs and monitors subject visit compliance (Business Layer).
- Trial Protocol & Candidate Data Store – persists protocol criteria, candidate, consent and compliance records (Data & Security Layer).
- Audit & Security Service – generates cryptographic audit trails for 21 CFR Part 11 compliance (Data & Security Layer).

### Interfaces (7 – exceeds the minimum of 4)

- `ISurveySubmission` – Candidate Portal (required) ↔ Eligibility Screening Engine (provided).
- `IComplianceReportView` – Researcher Dashboard (required) ↔ Visit Compliance Tracker (provided).
- `IProtocolCriteriaMgmt` – Researcher Dashboard (required) ↔ Trial Protocol & Candidate Data Store (provided).

- IQueryCriteria – Eligibility Screening Engine (required) ↔ Trial Protocol & Candidate Data Store (provided).
- IConsentRecordStore – Consent Management Service (required) ↔ Trial Protocol & Candidate Data Store (provided).
- IComplianceLogStore – Visit Compliance Tracker (required) ↔ Trial Protocol & Candidate Data Store (provided).
- «use» IAuditTrailWrite – usage dependency from Eligibility Screening Engine, Consent Management Service, and Visit Compliance Tracker to the Audit & Security Service.

## 2. Written Justification

We chose Layered Architecture for the Clinical Trial Subject Recruitment Engine.

### **Architectural Choice**

The system is organised into three horizontal layers: a Presentation Layer (Candidate Portal, Researcher Dashboard), a Business Logic Layer (Eligibility Screening Engine, Consent Management Service, Visit Compliance Tracker), and a Data & Security Layer (Trial Protocol & Candidate Data Store, Audit & Security Service).

### **Two Reasons**

- Regulatory workflow fit: candidate- and researcher-facing interfaces, protocol/eligibility/consent business rules, and record-keeping are naturally distinct concerns in a clinical-trial platform. Separating them into layers lets each concern evolve (e.g., updated trial protocols, new consent forms) without touching the UI or the storage/audit mechanisms.
- Consistent compliance logic: centralising trial-protocol data and the audit trail in one lower layer lets every business-layer service (eligibility engine, consent manager, compliance tracker) reuse the same validated data-access and logging path, instead of each service re-implementing its own compliance handling and risking inconsistent record-keeping.

### **Security Advantage**

Because every business-layer component must go through the Data & Security Layer to read or write trial data, cryptographic audit-trail generation required for 21 CFR Part 11 is enforced at a single choke point rather than left to individual services. This makes it far harder for any component to bypass audit logging or silently alter eligibility, consent, or compliance records.

### **Performance Benefit**

Because protocol criteria and candidate records are served from one well-defined data layer shared by all business-layer services, frequently accessed inclusion/exclusion criteria can be cached at that layer. This reduces redundant database round-trips when the Eligibility Screening Engine is evaluating many candidate submissions concurrently during an active recruitment drive.